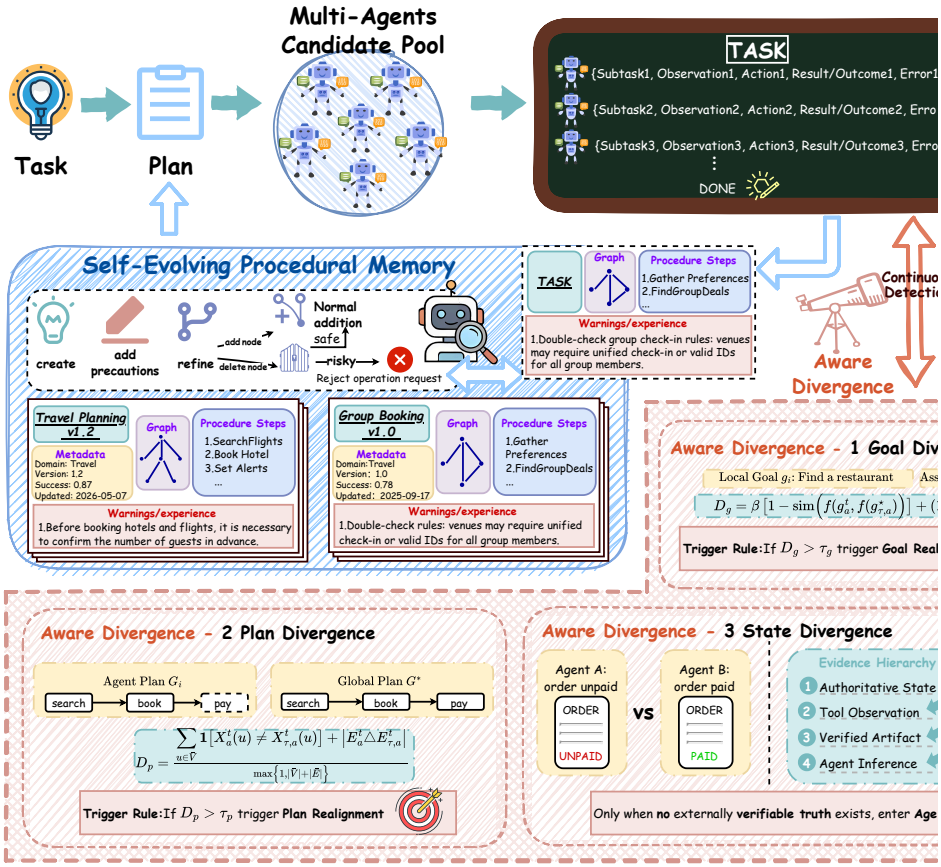
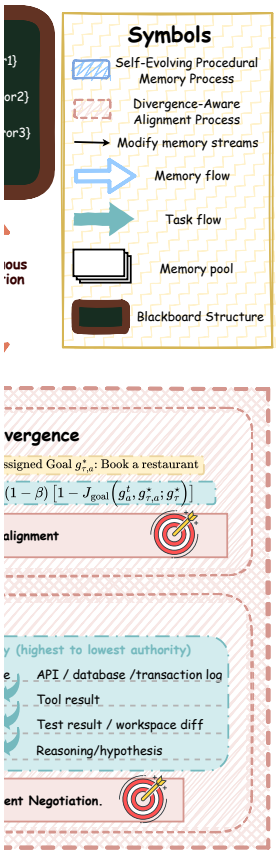

Multi-Agents Candidate Pool
Task
Plan
TASK
{Subtask1, Observation1, Action1, Result/Outcome1, Error1
{Subtask2, Observation2, Action2, Result/Outcome2, Erro
{Subtask3, Observation3, Action3, Result/Outcome3, Erro
DONE
Self-Evolving Procedural Memory
create
add precautions
refine
add node
delete node
Normal addition
safe
risky
Reject operation request
TASK
Graph
Procedure Steps
1.Gather Preferences
2.FindGroupDeals
Warnings/experience
1.Double-check group check-in rules: venues may require unified check-in or valid IDs for all group members.
Aware Divergence
Travel Planning v1.2
Metadata
Domain: Travel
Version: 1.2
Success: 0.87
Updated: 2026-05-07
Graph
Procedure Steps
1.SearchFlights
2.Book Hotel
3.Set Alerts
Warnings/experience
1.Before booking hotels and flights, it is necessary to confirm the number of guests in advance.
Group Booking v1.0
Metadata
Domain: Travel
Version: 1.0
Success: 0.78
Updated: 2025-09-17
Graph
Procedure Steps
1.Gather Preferences
2.FindGroupDeals
Warnings/experience
1.Double-check rules: venues may require unified check-in or valid IDs for all group members.
Aware Divergence - 1 Goal Div
Local Goal g_i: Find a restaurant
$D_g = \beta\left[1 - \mathrm{sim}\left(f(g_a^t, f(g_{\tau,a}^*)\right)\right] + ($
Trigger Rule:If $D_g > \tau_g$ trigger Goal Real
Aware Divergence - 2 Plan Divergence
Agent Plan G_i
search
book
pay
Global Plan G^*
search
book
pay
$D_p = \frac{\sum_{u \in \hat{V}} \mathbf{1}\left[X_a^t(u) \neq X_{\tau,a}^t(u)\right] + \left|E_a^t \triangle E_{\tau,a}^t\right|}{\max\left\{1, |\hat{V}| + |\hat{E}|\right\}}$
Trigger Rule:If $D_p > \tau_p$ trigger Plan Realignment
Aware Divergence - 3 State Divergence
Agent A: order unpaid
ORDER
UNPAID
vs
Agent B: order paid
ORDER
PAID
Evidence Hierarchy
1 Authoritative State
2 Tool Observation
3 Verified Artifact
4 Agent Inference
Only when no externally verifiable truth exists, enter Age


Symbols
Self-Evolving Procedural Memory Process
Divergence-Aware Alignment Process
Modify memory streams
Memory flow
Task flow
Memory pool
Blackboard Structure
vergence
ssigned Goal $g^*_{\tau,a}$: Book a restaurant
$(1-\beta)\left[1-J_{\text{goal}}\left(g^t_a, g^*_{\tau,a}; g^*_\tau\right)\right]$
alignment
y (highest to lowest authority)
API / database /transaction log
Tool result
Test result / workspace diff
Reasoning/hypothesis
ent Negotiation.